Mathematics

Equations and Percentages

Learning Material • Gymnasium Oberstufe

Overview

This comprehensive learning material covers fundamental mathematical concepts including **equivalent transformations** for solving equations and various **percentage calculations**. It includes practice problems from real-world scenarios, perfect for exam preparation and daily study.

Document Structure

- **PART 1: LEARNING** — Key concepts, definitions, and explained examples with step-by-step breakdowns
- **PART 2: PRACTICE** — Exercise questions with answers and detailed explanations

Key Features

- Bilingual content (English / German)
- Clear definitions and practical examples
- Exam-focused tips and tricks
- Real-world application problems

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How to Use This Material

Key Definitions

Pink boxes contain formal definitions, key theories, and foundational formulas to memorize.

Exam Tips

Yellow boxes highlight exam-focused tips, practical shortcuts, and problem-solving tricks.

PART 1: LEARNING

Understanding key concepts with detailed explanations

1. Equations

ENGLISH

This material covers fundamental concepts in mathematics, including solving equations and various types of percentage calculations. It also includes practice problems from real-world scenarios, suitable for high school (Gymnasium Oberstufe) level.

DEUTSCH

Dieses Material behandelt grundlegende mathematische Konzepte, einschliesslich des Lösens von Gleichungen und verschiedener Arten von Prozentrechnungen. Es enthält auch Übungsaufgaben aus realen Szenarien, geeignet für das Gymnasium (Oberstufe).

1.1 Equivalent Transformations

Definition — Equation

An equation consists of a right term and a left term, connected by an equality sign (=). The general form is:

$$T_L = T_R$$

Where T_L is the left term and T_R is the right term. In elementary algebra, equations typically involve one unknown variable that needs to be solved for.

Definition — Equivalent Transformations

Equivalent transformations are operations performed on both sides of an equation that do not change the solution set. These operations are essential for isolating the unknown variable.

1.1.1 Addition

ENGLISH

To solve an equation by isolating 'x', we perform the inverse operation on both sides.

Problem:

x + 3 = 7

Operation:

DEUTSCH

Um eine Gleichung durch Isolieren von 'x' zu lösen, führen wir die umgekehrte Operation auf beiden Seiten durch.

Problem:

x + 3 = 7

Operation:

Subtract 3 from both sides.

```
x + 3 - 3 = 7 - 3
x = 4
```

Also:

3 + x = 7 → x = 7 - 3 → x = 4

Subtrahiere 3 von beiden Seiten.

```
x + 3 - 3 = 7 - 3
x = 4
```

Auch:

3 + x = 7 → x = 7 - 3 → x = 4

1.1.2 Subtraction

ENGLISH

To solve an equation where a number is subtracted from 'x', we add that number to both sides.

Problem:

x - 3 = 7

Operation:

Add 3 to both sides.

```
x - 3 + 3 = 7 + 3
x = 10
```

DEUTSCH

Um eine Gleichung zu lösen, bei der eine Zahl von 'x' subtrahiert wird, addieren wir diese Zahl zu beiden Seiten.

Problem:

x - 3 = 7

Operation:

Addiere 3 zu beiden Seiten.

```
x - 3 + 3 = 7 + 3
x = 10
```

1.1.3 Multiplication

ENGLISH

To solve an equation where 'x' is multiplied by a number, we divide both sides by that number.

Problem:

$x \cdot 3 = 11$

Operation:

Divide both sides by 3.

$$x = 11 : 3$$
$$x = 11/3$$

Also:

$3 \cdot x = 11 \rightarrow x = 11/3$

DEUTSCH

Um eine Gleichung zu lösen, bei der 'x' mit einer Zahl multipliziert wird, dividieren wir beide Seiten durch diese Zahl.

Problem:

$x \cdot 3 = 11$

Operation:

Dividiere beide Seiten durch 3.

$$x = 11 : 3$$
$$x = 11/3$$

Auch:

$3 \cdot x = 11 \rightarrow x = 11/3$

1.1.4 Division

ENGLISH

To solve an equation where 'x' is divided by a number, we multiply both sides by that number.

Problem:

$x : 3 = 11$

Operation:

Multiply both sides by 3.

$$x = 11 \cdot 3$$
$$x = 33$$

Also:

$x/3 = 11 \rightarrow x = 11 \cdot 3 \rightarrow x = 33$

DEUTSCH

Um eine Gleichung zu lösen, bei der 'x' durch eine Zahl dividiert wird, multiplizieren wir beide Seiten mit dieser Zahl.

Problem:

$x : 3 = 11$

Operation:

Multipliziere beide Seiten mit 3.

$$x = 11 \cdot 3$$
$$x = 33$$

Auch:

$x/3 = 11 \rightarrow x = 11 \cdot 3 \rightarrow x = 33$

LEARNING EXERCISE 1: SOLVING EQUATIONS

Problem: Solve for x: $4x + 7 = 23$

> **Step 1:** Identify what needs to be isolated. We want to get x alone on one side of the equation.

> **Step 2:** The term "4x" means 4 times x. First, remove the +7 from the left side by doing the opposite operation — subtraction.

> **Step 3:** Subtract 7 from BOTH sides:
>
> $$4x + 7 - 7 = 23 - 7$$
> $$4x = 16$$

> **Step 4:** Now we have $4x = 16$. To isolate x, divide by 4 (the opposite of multiplication):
>
> $$4x \div 4 = 16 \div 4$$
> $$x = 4$$

> **Step 5 (Verification):** Always check your answer by substituting back:
>
> $$4(4) + 7 = 16 + 7 = 23 \checkmark$$

Try it yourself: Solve $5x - 3 = 22$

2. Percentage Calculation

ENGLISH

Percentage calculation is a fundamental skill in mathematics, used to express parts of a whole or changes in quantities.

DEUTSCH

Die Prozentrechnung ist eine grundlegende mathematische Fähigkeit, die verwendet wird, um Teile eines Ganzen oder Änderungen von Mengen auszudrücken.

2.1 Decimal Conversion

ENGLISH

To perform calculations with percentages, it is often necessary to convert percentages into decimal numbers.

> **Important Information**
>
> To convert a percentage to a decimal, divide the percentage value by 100. This is equivalent to moving the decimal point two places to the left.

Examples:

- 0.3% = 0.3 ÷ 100 = `0.003`
- 12% = 12 ÷ 100 = `0.12`

DEUTSCH

Um Berechnungen mit Prozenten durchzuführen, ist es oft notwendig, Prozentsätze in Dezimalzahlen umzuwandeln.

> **Wichtige Information**
>
> Um einen Prozentsatz in eine Dezimalzahl umzuwandeln, dividieren Sie den Prozentwert durch 100.

Beispiele:

- 0.3% = 0.3 ÷ 100 = `0.003`
- 12% = 12 ÷ 100 = `0.12`

2.2 Terminology: "Um" (By) vs. "Auf" (To)

ENGLISH

In German, the prepositions "um" (by) and "auf" (to) are crucial for understanding percentage changes.

- **"Um" (By):** Refers to the *change amount*. "Es steigt um 20%" means an addition of 20% to the original value.
- **"Auf" (To):** Refers to the *new total value* as a percentage of the original. "Es steigt auf 120%" means the new value is 120% of the original.

DEUTSCH

Im Deutschen sind die Präpositionen „um" und „auf" entscheidend für prozentuale Veränderungen.

- **„Um" (By):** Bezieht sich auf die *Veränderung*.
- **„Auf" (To):** Bezieht sich auf den *neuen Gesamtwert*.

Exam Tip

When calculating, convert "auf" percentages directly into a multiplication factor. For "um" percentages, first determine the "auf" percentage.

Example:

Price increases by 20%

$$P_{neu} = P_{alt} \cdot (1 + 0.2)$$
$$P_{neu} = P_{alt} \cdot 1.2$$

Prüfungstipp

Wandeln Sie „auf"-Prozentsätze direkt in einen Multiplikationsfaktor um.

Beispiel:

Preis steigt um 20%

$$P_{neu} = P_{alt} \cdot (1 + 0.2)$$
$$P_{neu} = P_{alt} \cdot 1.2$$

2.3 Key Components of Percentage Calculation

Definition — Key Components

A: Anteil (Part)
The value after a change or the specific part of the base value.

G: Grundwert (Base)
The original value before a change.

P: Prozentsatz (Rate)
The percentage expressed as a decimal (e.g., 20% = 0.2).

$$A = G \cdot P$$

2.4 Types of Percentage Calculations

2.4.1 Anteilberechnung (Calculating the Part)

ENGLISH

Goal: Find A when G and P are known.

Formula: $A = G \cdot P$

Problem:

A fruit farmer harvested 960 kg of apples, 80% sold in farm shop. How many kg sold?

Given: G = 960 kg, P = 80% = 0.8

Calculation: A = 960 · 0.8 = 768 kg

Result: 768 kg

DEUTSCH

Ziel: Finde A wenn G und P bekannt sind.

Formel: $A = G \cdot P$

Aufgabe:

960 kg Äpfel, 80% verkauft.

Gegeben: G = 960 kg, P = 0.8

Berechnung: A = 960 · 0.8 = 768 kg

Ergebnis: 768 kg

2.4.2 Grundwertberechnung (Calculating the Base Value)

ENGLISH

Goal: Find G when A and P are known.

Formula: G = `A/P`

Problem:

5 students are sick = 20% of class. Total students?

DEUTSCH

Ziel: Finde G wenn A und P bekannt sind.

Formel: G = `A/P`

Aufgabe:

5 Schüler = 20% der Klasse.

Gegeben: A = 5, P = 0.2

Given: A = 5, P = 20% = 0.2

Calculation: G = `5/0.2` = 25

Result: 25 students

Berechnung: G = `5/0.2` = 25

Ergebnis: 25 Schüler

2.4 Change Factor (Änderungsfaktor)

Important Information

Using a change factor (also called a growth factor or multiplier) simplifies calculations involving percentage increases or decreases.

- **Increase by p%:** Multiply by (1 + p/100)
- **Decrease by p%:** Multiply by (1 - p/100)

ENGLISH

Example: USB-Stick Price Increase

Problem:

USB stick costs €12. Price increases by 4%.

Solution:

Change factor = 1 + 4/100 = 1.04

New price = 12 · 1.04 = €12.48

Result: €12.48

Example: Capri Pants Price Decrease

Problem:

Pants cost €64. Price reduced by 25%.

Solution:

Change factor = 1 - 25/100 = 0.75

Reduced price = 64 · 0.75 = €48

Result: €48

DEUTSCH

Beispiel: USB-Stick Preiserhöhung

Aufgabe:

USB-Stick kostet €12. Preis +4%.

Lösung:

Änderungsfaktor = 1 + 4/100 = 1.04

Neuer Preis = 12 · 1.04 = €12.48

Ergebnis: €12.48

Beispiel: Caprihose Preisreduktion

Aufgabe:

Hose kostet €64. Preis -25%.

Lösung:

Änderungsfaktor = 1 - 25/100 = 0.75

Preis = 64 · 0.75 = €48

Ergebnis: €48

LEARNING EXERCISE 2: UNDERSTANDING "UM" VS "AUF"

Scenario: A product costs €100. Answer these two questions:

> **Question A:** The price *increases by 30%*. What is the new price?
> **Analysis:** "Um 30%" means the change is 30% of the original. The new price will be 100% + 30% = 130% of the original.
>
> ```
> New price = €100 × 1.30 = €130
> ```

> **Question B:** The price *increases to 130%* of the original. What is the new price?
> **Analysis:** "Auf 130%" directly tells us the new price as a percentage of the original. This is the same as increasing by 30%!
>
> ```
> New price = €100 × 1.30 = €130
> ```

> **Key Insight:** Both questions give the same answer! The difference is:
>
> - "Um 30%" → tells you the *change* (the increase)
> - "Auf 130%" → tells you the *result* (the final percentage)

Try it: If €50 increases BY 20%, what's the new price? If €50 increases TO 120%, what's the new price?

2.5 Percentage Points (Prozentpunkte)

Definition — Percentage Points

A percentage point is a unit used to express the arithmetic difference between two percentages. It represents an absolute difference, unlike a percentage change which is a relative difference.

Note

Percentage points are commonly used in statistics, especially when comparing election results, interest rates, or survey data, to avoid ambiguity.

ENGLISH

Example: Interest Rate Increase

Problem:

Interest rate increases from 5% to 6%.

a) Percentage point increase: 6% - 5% = **1 percentage point**

b) Percentage increase: (6%-5%)/5% = $1/5$ = 0.2 = **20%**

> The rate increased by 1 pp, which is a 20% relative increase.

DEUTSCH

Beispiel: Zinssatzerhöhung

Aufgabe:

Zinssatz steigt von 5% auf 6%.

a) Prozentpunkt-Erhöhung: 6% - 5% = **1 Prozentpunkt**

b) Prozentuale Erhöhung: (6%-5%)/5% = $1/5$ = 0.2 = **20%**

> Der Satz stieg um 1 PP, was 20% relative Erhöhung ist.

Exam Tip

When comparing percentages in exams, always clarify whether you're talking about:

- **Percentage points** — the simple difference (e.g., from 40% to 50% = +10 pp)
- **Percentage change** — the relative change (e.g., from 40% to 50% = +25%)

2.6 Visual: Water Distribution on Earth

ENGLISH

The graphic illustrates the total water distribution on Earth.

- **Global Water Volume:** 1,317,398,571 km³ of seawater (97%)
- **Usable Water:** Only surface water and groundwater are usable for drinking water.

DEUTSCH

Die Grafik stellt die Gesamtwasserverteilung der Erde dar.

- **Gesamtwasser:** 1.317.398.571 km³ Meerwasser (97%)
- **Nutzbares Wasser:** Nur Oberflächen- und Grundwasser sind nutzbar.

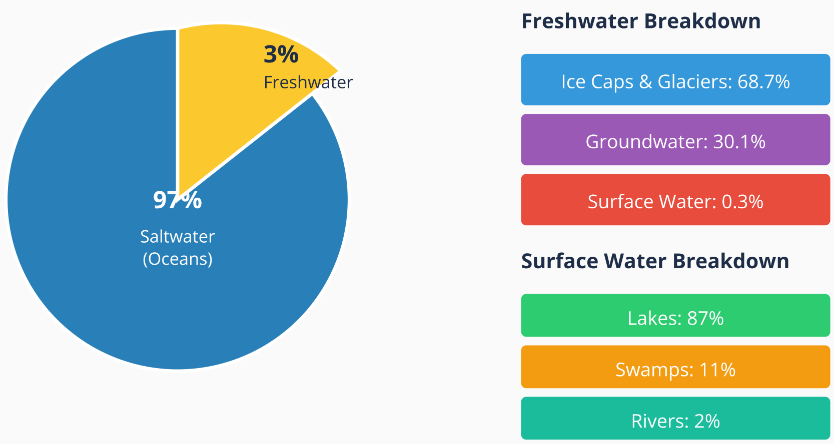

Figure 1: Global Water Distribution on Earth

Key Insight

Notice that only 0.3% of all water is surface water (lakes, rivers, swamps), and of that tiny amount, 87% is in lakes. This means lakes represent only about 0.26% of all Earth's water!

PART 2: PRACTICE

Exercise questions with detailed answers and explanations

3. Practice: Equations

Question 1

Solve for x: $3x + 8 = 29$

Answer: $x = 7$

Step-by-step breakdown:
1. Start with: $3x + 8 = 29$
2. Subtract 8 from both sides: $3x = 29 - 8 = 21$
3. Divide both sides by 3: $x = 21 \div 3 = 7$
4. Check: $3(7) + 8 = 21 + 8 = 29$ ✓

Question 2

Solve for x: $x/4 - 5 = 3$

Answer: $x = 32$

Step-by-step breakdown:
1. Start with: $x/4 - 5 = 3$
2. Add 5 to both sides: $x/4 = 3 + 5 = 8$
3. Multiply both sides by 4: $x = 8 \times 4 = 32$
4. Check: $32/4 - 5 = 8 - 5 = 3$ ✓

Question 3

Solve for x: $2(x - 3) = 14$

Answer: $x = 10$

Step-by-step breakdown:

1. Start with: $2(x - 3) = 14$
2. Divide both sides by 2: $x - 3 = 14 \div 2 = 7$
3. Add 3 to both sides: $x = 7 + 3 = 10$
4. Check: $2(10 - 3) = 2 \times 7 = 14$ ✓

Why this works: We use the distributive property in reverse. Dividing first is easier than expanding $2x - 6 = 14$.

4. Practice: Percentage

Question 1

A laptop costs €1,450 including 20% VAT. What is the net price (without VAT)?

Answer: €1,208.33

Step-by-step breakdown:
Understanding: The gross price (€1,450) = 120% of the net price (100% + 20% VAT)

1. Let G = net price (what we're looking for)
2. We know: G × 1.20 = €1,450
3. Therefore: G = €1,450 ÷ 1.20 = €1,208.33

Why this works: To remove VAT, we divide by 1.20 (the "auf" factor). To add VAT, we multiply by 1.20.

Question 2

A car's value decreases from €25,000 to €18,500. What is the percentage decrease?

Answer: 26% decrease

Step-by-step breakdown:
1. Calculate the decrease: €25,000 - €18,500 = €6,500
2. Find what percentage €6,500 is of the original €25,000:
3. Percentage = (6,500 ÷ 25,000) × 100 = 0.26 × 100 = 26%

Formula: Percentage decrease = (Original - New) ÷ Original × 100

Question 3

Shop A offers 20% discount, then additional 15% off the sale price. Shop B offers 35% discount on the original price. Which shop is better?

Answer: Shop A is better (68% vs 65% of original)

Step-by-step breakdown:
Shop A (sequential discounts):
- First discount: 100% - 20% = 80% remaining
- Second discount: 80% × (100% - 15%) = 80% × 85% = 68%

Shop B (single discount):
- 100% - 35% = 65% remaining

Why this works: Sequential discounts are calculated on the *reduced* amount, not the original. This is why Shop A leaves you paying less!

Question 4

An interest rate changes from 4% to 5%. Express this as: (a) percentage points, (b) percentage change.

Answer: (a) 1 percentage point (b) 25% increase

Step-by-step breakdown:
(a) Percentage points: 5% - 4% = **1 pp**
(b) Percentage change: (5% - 4%) ÷ 4% = 1% ÷ 4% = 0.25 = **25%**

Key distinction: The rate went UP by 1 percentage point, but RELATIVELY it increased by 25% (of the original 4%).

5. Practice: Applications

Question 1

Sarah buys a computer for €1,600. The price first increases by 10%, then decreases by 10%. What is the final price?

Answer: €1,584

Step-by-step breakdown:
1. Increase by 10%: €1,600 × 1.10 = €1,760
2. Then decrease by 10%: €1,760 × 0.90 = €1,584

Why this is tricky: The 10% decrease is calculated on €1,760 (the increased amount), not the original €1,600!

Quick method: Multiply both factors: 1.10 × 0.90 = 0.99
Final = €1,600 × 0.99 = €1,584

Key insight: A 10% increase followed by a 10% decrease does NOT get you back to the original! You lose 1%.

Question 2

At an election, Party A gained 45% of votes (up from 36% last time). How many percentage points did they gain? What was their percentage increase?

Answer: 9 percentage points, 25% increase

Step-by-step breakdown:
1. Percentage points: 45% - 36% = **9 pp**
2. Percentage change: (45% - 36%) ÷ 36% = 9% ÷ 36% = 0.25 = **25%**

Why both matter: - "9 percentage points gained" tells you the absolute difference in the score
- "25% increase" tells you the relative growth from their previous position

Question 3

Using the water distribution data, calculate what percentage of ALL water on Earth is surface water (lakes + swamps + rivers)?

Answer: 0.00783%

Step-by-step breakdown:
From the diagram:
- Saltwater: 97% = 0.97
- Freshwater: 3% = 0.03
- Surface water (of freshwater): 0.3% = 0.003

Calculation: 0.03 × 0.003 = 0.0000783
As percentage: 0.0000783 × 100 = **0.00783%**

Why this matters: This tiny percentage explains why freshwater is such a precious resource!